

3.5.2 Selection Statements: if and if-else

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 15 of 24

★ Golden

3.5.2 Selection Statements: if and if-else ★ Golden

LEARN IT

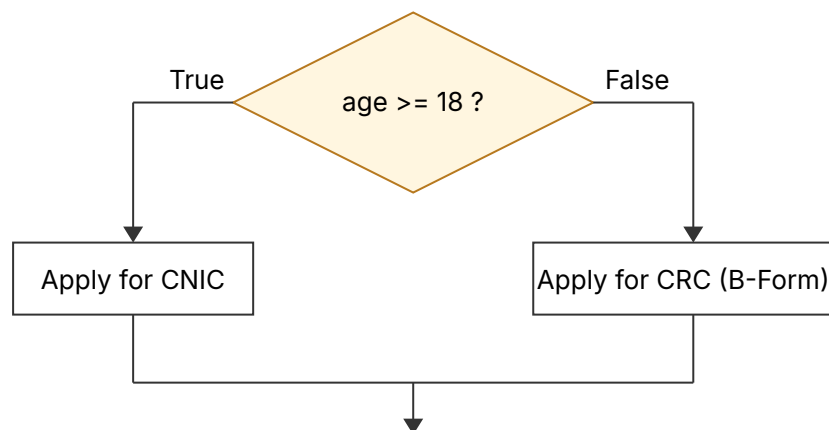
Selection changes the flow of a program based on a **condition** — an expression that is either True or False. It helps a program respond intelligently to different situations.

1. The if statement executes a block of code **only if** the condition is True. If the condition is False, the block is completely skipped.

```
if condition:
    statement(s)    # runs only when condition is True
```

2. The if-else statement provides **two paths**: if the condition is true, do this; **else** (otherwise), do that. Exactly one of the two blocks always runs.

```
if condition:
    block A         # condition True
else:
    block B         # condition False
```



An if-else statement: one diamond (condition), two paths, then the paths join again.

WORKED EXAMPLE

The **if** statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("You are an Adult.")    # runs only if age ≥ 18
print("Program continues...")    # runs every time
```

Run 1 (user types 20):

```
OUTPUT
Enter your age: 20
You are an Adult.
Program continues...
```

Run 2 (user types 15):

```
OUTPUT
Enter your age: 15
Program continues...
```

The last line is not indented, so it is outside the `if` block and runs every time.

The if-else statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("Apply for CNIC.")
else:
    print("Apply for CRC (B-Form).")
```

Sample run (user types 16):

```
OUTPUT
Enter your age: 16
Apply for CRC (B-Form).
```

Even or odd — a classic if-else program:

```
n = 7
if n % 2 == 0:
    print(n, "is even")
else:
    print(n, "is odd")
```

```
OUTPUT
7 is odd
```

$7 \% 2 = 1$, and $1 == 0$ is False, so the `else` block runs.

REMEMBER IT

Formatting rule: put a **colon (:)** after `if` and `else`, and **indent** the code inside the blocks. `else` never has a condition of its own.

COMMON MISTAKE

Common mistakes: forgetting the colon (`SyntaxError: expected ':'`), writing a condition after `else` (`else age < 18: is wrong`), and not indenting `else` at the same level as its `if`.

اردو میں سمجھیں

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔ `if` کا بلاک صرف تب چلتا ہے جب شرط درست (True) ہو۔ `if-else` میں دو راستے ہوتے ہیں: شرط درست ہو تو پہلا بلاک، ورنہ `else` والا بلاک چلتا ہے۔ `if` اور `else` کے بعد کولن (:) اور اندر کے کوڈ کی انڈینٹیشن ضروری ہے۔

EXAM TIP

Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`, the condition, both blocks, and a sample run.

CHECK YOURSELF

1. Predict the output when `marks = 32`:

```
marks = 32
if marks ≥ 33:
    print("Pass")
else:
    print("Fail")
print("Done")
```

2. Find the two errors:

```
n = 5
if n > 0
    print("Positive")
else n ≤ 0:
    print("Not positive")
```

Answers: 1. Fail then Done . 2. Missing colon after `if n > 0`; `else` must not have a condition — write `else: .`

Next lecture: 3.5.2 Selection Statements: if-elif-else and Nested if